

SolaX Power Battery Storage Requirements



Storage & Battery

Lithium is a particularly susceptible metal to chemical reactions and is prone to extension and combustion. In the lithium battery packaging and transport, if carried out improper handling, it is easy to lead to combustion and explosion, and accidents have occurred from time to time. Therefore, the storage management of lithium batteries is required as follows:

1. Storage requirements

- a) Due to the characteristics of lithium batteries, the storage environment temperature needs to meet the following requirements:

Battery Type	T-BAT-SYS-HV-4.5/6.3	T-BAT-SYS-HV-5.8	T-BAT-SYS-HV-3.0 T-BAT-SYS-HV-R2.5/3.6 T-BAT-SYS-HV-5.0 T-BAT-SYS-HV-S2.5/3.6 T-BAT-SYS-LV-R2.5/3.6 T-BAT-SYS-HV-S50E T-BAT-SYS-LV D53
Storage Temperature	-20 ℃ ~ 45 ℃	-20 ℃ ~ 30 ℃	-20 ℃ ~ 30 ℃
	0 ℃ ~ 40 °	30 ℃ ~ 55 °	30 ℃ ~ 50 ℃

- b) Control humidity in the warehouse effectively, avoiding extreme humidity for a long time. The relative humidity range should be between 40% and 60%RH.

- c) Lithium battery warehouses should be physically isolated by brick walls, and the warehouses must be enclosed, explosion-proof and equipped with other corresponding safety electrical lighting equipment.

- d) Locations where flammable and explosive materials are stored should be equipped with a sufficient variety and quantity of fire-fighting equipment, which shall always be in good condition.

- e) Where there are lithium batteries, there must be some safety regulations such as no smoking.

f) It is forbidden to carry out operations that may cause fire in warehouses and sites where flammable and explosive materials are stored.

g) Lithium batteries should be stored in a well-ventilated, dry and cool place. High temperature and high humidity may damage the battery performance and/or corrode the battery surface.

h) Battery cartons should not be stacked higher than the specified height. If too many battery cartons are stacked together, the batteries in the bottom cartons may be deformed, which may lead to liquid leakage.

i) Avoid storing or displaying the battery in direct sunlight or in areas exposed to rain. A rise in temperature may damage the battery. When batteries are exposed to rain, the insulation resistance will decrease, and self-discharge and rust may occur.

j) Store and display batteries in their original packaging. Stacking unpacked batteries indiscriminately is likely to cause short-circuiting and damage to the batteries.

k) Items that are likely to cause combustion or explosion when they come into contact with each other and items with different fire-fighting methods should be stored separately.

2. Recharge requirements

a) Proper recharge maintenance of batteries can substantially improve their performance and extend their service life. At different storage temperatures, the battery needs to be recharged according to the following different requirements:

Battery Type	Maintenance
T-BAT-SYS-HV-4.5/6.3	-20 °C ~ 45 °C, recharge the batteries at least once every 3 months
	0 °C ~ 40 °C, recharge the batteries at least once every 1 year
T-BAT-SYS-HV-5.8	-20 °C ~ 30 °C, recharge the batteries at least once every 1 year
	30 °C ~ 55 °C, recharge the batteries at least once every 6 months
T-BAT-SYS-HV-3.0	-20 °C ~ 30 °C, recharge the batteries at least once every 1 year
T-BAT-SYS-HV-R2.5/3.6	
T-BAT-SYS-HV-5.0	

T-BAT-SYS-HV-S2.5/3.6 T-BAT-SYS-LV-R2.5/3.6 T-BAT-SYS-HV-S50E T-BAT-SYS-LV D53	30 ℃ ~ 50 ℃, recharge the batteries at least once every 6 months
---	--

b) If there is a label on the battery carton stating the last charging time, please use this time as the starting time and then recharge the battery with the above listed frequency. If there is no label on the battery carton stating the last charge time, please contact sales or service support team to obtain the last recharge time of the battery.

c) For specific recharge operations, please refer to T-BAT-Charger Installation Manual.

d) Collecting the time information of recharge operations carried out after the battery putting into storage.

3. Handling requirements

a) Handlers should use qualified handling tools (forklift, trolley, etc.).

b) When handling materials, no matter what kind of handling tools are used, they should be handled properly to prevent materials from falling or being damaged.

c) When handling materials, load, stacking, directionality and other issues should be considered.

d) When using forklifts or carts, the loading method should be heavy and light, and the material to be transported should not exceed the width and height of the passage and elevator doors, and the speed should be moderate.

e) The battery cartons should be handled carefully. Rough handling may cause short circuit or damage to the battery, resulting in leakage, explosion or fire.

4. Processing requirements

- a) Only in the case of consistent with local regulations, lithium (primary) batteries can be used for public garbage disposal.
- b) Do not disassemble the battery.
- c) Except for the use of approved controllable stoves, it is forbidden to dispose of batteries by fire.

5. Warehouse requirements

- a) Warehouse management personnel should check the cargo information every day. If they find that the storage location is incorrect, the accounts are inconsistent, the quality problems should be reported and dealt with in a timely manner, they should conduct a fire prevention inspection and cut off the power supply at the end of work or after get off work.
- b) Clean up the warehouse area every day, clean up dirt and debris on the ground in time, and arrange the materials in the warehouse into the designated area to meet the requirements of neat, tidy, clean, hygienic and reasonable placement.
- c) Keep safe passages in the warehouse unblocked to prevent accumulations and ensure the safety of personnel.
- d) The planning area in the warehouse must be clearly marked, and the material storage area must be classified and stored in different districts, and clearly marked.

6. Out of warehouse requirements

- a) The battery packs are shipped on a first-in-first-out basis.

b) Use the same batch group together, and try not to use different batches in combination. Otherwise, it will affect the service life of the overall system and cause the after-sales problem of battery imbalance. (Note: T-BAT-SYS-LV-R2.5/3.6 and T-BAT-SYS-LV D53 are not affected by this requirement.)

c) Collecting the time when batteries are put into storage. If different batches of batteries are shipped, and the interval between batches and batches of storage time is greater than or equal to 3 months, it is necessary to recharge the batteries in this group to the same SOC before shipment. This is to prevent uneven battery capacity when the batteries arrive at the user's end.

7. Emergency treatment methods

Lithium batteries may leak, rust and expand when stored for a long time; if they are not handled properly, they may heat up, burn or explode. The relevant treatment methods are as follows:

a) Rust treatment method: usually occurs in cylindrical lithium battery (polymer lithium battery does not have this problem), in the beginning slight rust will not affect the performance of the lithium battery, and it can be used normally. If the rust is serious (such as the part of the cap), it will affect the sealing performance of the battery and leak, and must be scrapped.

b) Treatment of leakage or swelling: leakage refers to the leakage of electrolyte in the battery, which usually has a pungent odor. The strong corrosiveness of the electrolyte will cause damage to the battery protection board components, such as polymer Lithium batteries will swell. Leaking and swollen batteries must be sorted out and discarded.

c) Under normal environmental conditions of temperature and humidity, the battery will not produce mildew or discoloration. If leakage occurs, such undesirable phenomena will flourish.

d) Heat treatment method: it will generate heat during charging and discharging (under use), but the temperature is usually below 60 degrees. The temperature of the battery can reach

hundreds or hundreds of hundreds in the state of internal or external short circuit. At this time, the battery must be isolated and placed in the sand. Be careful not to touch the battery directly with your hands, otherwise it will be burned. After the battery temperature drops to normal temperature, discard it.

e) Combustion or explosion treatment method: If combustion or explosion occurs, it is very dangerous, and personnel must stay away; battery burning will not produce open flames, usually due to high levels of ignition of nearby flammable products or packaging outer boxes, take protective measures Under the premise, such as the fire extinguishing method of a single or very few batteries burning or exploding; use sand to directly cover the burning or exploding battery. If it is a large area of burning, use a dry powder fire extinguisher to extinguish the fire. It is strictly forbidden to use water to extinguish the fire, because water will cause the battery to short circuit.